O-B-420B
July 22, 1970
SUPERSEDING
Fed. Spec. O-B-420A
December 2, 1964

FEDERAL SPECIFICATION

BLEACH, LAUNDRY, ORGANIC CHLORINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of organic chlorine, dry laundry bleach (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

P-D-245	-	Detergent (Synthetic) Built, High and Low Sudsing.
PPP-B-26	-	Bag, Plastic, Polyethylene.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-D-723	-	Drums, Fiber.
PPP-T-60	-	Tape; Pressure Sensitive Adhesive, Waterproof for Packaging.
PPP-T-76	-	Tape, Pressure Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 123	-	Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536	-	Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

FSC 6850

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

U.S. Department of Health, Education, and Welfare Publication:

21CFR 191 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are on sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 2022-64 - Standard Methods of Sampling and Chemical Analysis of Chlorine Containing Bleaches.
E 70-68 - Standard Method of Test for pH of Aqueous Solutions With the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The laundry bleach shall be a white product in powder or bead form containing an organic, chlorine-liberating compound. The organic bleach shall not be compounded using trichloroisocyanuric acid, or mixtures containing trichloroisocyanuric acids. It shall not contain caustic soda.

3.2 Chemical and physical requirements. The product shall conform to the requirements specified in table I when tested as specified in 4.3.4.

TABLE I. Requirements

	Minimum	Maximum
Available chlorine, percent	15.0	17.0
Phosphates, as P_2O_5, percent 1/	14.0	-
Carbonates, as CO_2, percent	-	0.2
Reaction with carbonate ion	slight turbidity	
pH of an aqueous solution	7.8	9.0
Insoluble matter content, percent 2/	-	1.5
Moisture content, percent	-	2.0

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TABLE I. Requirements (cont'd)

	Minimum	Maximum
Loss on ignition, percent	18.0	-
Thermal decomposition point, degrees centigrade	200	-
Loss of available chlorine on heating, percent	-	20.0
Particle size, pass through: 16 mesh sieve, percent	90	-
100 mesh sieve, percent	-	20
200 mesh sieve, percent	-	5

1/ The phosphate content shall be made up of sodium tripolyphosphate or tetrasodium pyrophosphate or a mixture of these.

2/ To include dedusting agent, if present.

3.3 Performance. The bleach shall remove not less than 78 percent of the stain when tested as specified in 4.3.4.

3.4 Workmanship. The material shall be clean, uniform, free flowing, shall show no evidence of segregation of individual constituents or of lumping and caking and no foreign material shall be present.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents. The supplier shall submit a certificate of compliance satisfying that the phosphates are made up of sodium tripolyphosphate or tetrasodium pyrophosphate or a mixture of these. The certificate shall be accompanied by actual test, inspection or other verifiable quality data.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs and at the inspection levels and the acceptable quality levels (AQL's) set forth in 4.3.3. Each lot of end items shall be examined for visual, contents, and preparation for delivery defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity for the examinations in 4.3.1.1 and 4.3.1.2, in units of shipping containers for the examination in 4.3.2.1.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Container:	
Construction	Not as specified (see 5.1 and 5.2).
Workmanship	Dented, crushed, cracked, abraded, dirty, or torn.
Closure	Incomplete, imperfect, leakage or damage to polyethylene liner.
Label	Missing, incomplete, incorrect. Not as specified (see 5.3.3).
Liner	Missing. Not as specified; punctured or damaged. Not completely heat sealed at seams, knotted, or not double folded and taped (as applicable).
Packing	Not in accordance with requirements (see 5.2).

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Examine	Defect
Contents:	
Appearance	Presence of foreign material. Not uniform Not free flowing. Contains agglomerates, cakes, or lumps. Not clean.

4.3.1.2 Examination of filled containers for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container, for all sample units examined, is less than required.

4.3.2 Examination of preparation for delivery.

4.3.2.1 Examination for packaging, packing, and marking. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit for this examination shall be one shipping container fully prepared for delivery and need not be sealed. Shipping containers fully prepared for delivery shall be examined for defects of closure. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Markings	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components: such as incomplete container flaps; loose or inadequate sealing, strapping, or stapling. Bulged or distorted containers.
Contents	Number of packages (where applicable) per shipping container is more or less than required. Container (where applicable) not as specified.

4.3.3 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A
4.3.2.1	S-2	2.5

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4.3.4 Testing of the end item. The method of test specified in Fed. Test Method Std. No. 536 whenever applicable and as listed in table II shall be followed. The sample unit for testing shall be as specified in 4.3.4.1 and 4.3.4.2. The lot size shall be expressed in pounds of bleach. The sample size shall be as shown below. When specified in table II, tests shall be performed on a composite sample or sample unit basis. The lot shall be unacceptable if one or more sample units or the composite sample fail to meet any test requirements specified.

4.3.4.1 The sample for composite testing shall be two pounds of bleach, selected in equal portions in the frequency in 4.3.4.2. The portions from the individual samples shall be thoroughly mixed, yielding the composite. The sample shall be placed in an airtight and watertight brown glass container with suitable caps, covered, and sealed, to prevent any moisture change.

4.3.4.2 The sample unit for testing on a sample unit basis shall be 5 ounces, drawn in segregated 1-ounce portions and placed into 5 individual 1-ounce brown glass airtight and watertight jars, covered, and sealed to prevent any moisture change. A separate 1-ounce portion shall be used for each test made on a sample unit basis.

Lot size (pounds)	Sample size or frequency (No. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE II. Instructions for testing of the end item

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Individ. unit	Requirements applicable to: Composite sample	Number of determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Material	3.1	1/					
Available chlorine, percent	Table I	4.4.1	X		Aver. of 2		0.1
P_2O_5 content, percent	Table I	1501 2/		X	Aver. of 2		0.1
CO_2, content, percent	Table I	4.4.2		X	Aver. of 2		0.01

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TABLE II. Instructions for testing of the end item (cont'd)

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Individ. unit	Requirements applicable to: Composite sample	Number of determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Reaction with carbonate ion, turbidity, or precipitation	Table I	4.4.3		X	Aver. of 2	X	
pH of an aqueous solution	Table I	4.4.4		X	Aver. of 2		0.1
Insoluble matter content, percent	Table I	4.4.5	X		Aver. of 2		0.1
Moisture content, percent	Table I	4.4.6	X		Aver. of 2		0.1
Loss of ignition	Table I	4.4.7		X	Aver. of 3		0.1
Thermal decomposition point, degrees centigrade	Table I	4.4.8		X	Aver. of 2		1
Loss of available chlorine on heating, percent	Table I	4.4.9	X		Aver. of 2		0.1

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TABLE II. Instructions for testing of the end item (cont'd)

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Individ. unit	Requirements applicable to: Composite sample	Number of determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Particle size: Pass through 16 mesh sieve, percent	Table I	2101 2/		X	1		1.0
100 mesh sieve, percent	Table I	2101 2/		X	1		1.0
200 mesh sieve, percent	Table I	2101 2/		X	1		1.0
Performance, percent	3.3	4.4.10		X	1		1.0

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for requirement stated.

2/ Reference Fed. Test Method Std. No. 536.

4.4 Test procedures. Bleach sample stored in brown glass containers shall be tightly sealed and shall not be exposed to sunlight or to the atmosphere except when necessary during the course of analysis. Prior to each analysis, except where otherwise specified, a sample of bleach, approximately double the weight required shall be thoroughly ground and mixed in a dry mortar. The sample required for the analysis shall be immediately weighed from the ground sample.

4.4.1 Available chlorine.

4.4.1.1 Reagents.

a. Sodium thiosulfate ($Na_2S_2O_3$), ACS grade.
b. Hydrochloric acid (HCl), ACS grade.
c. Potassium iodide (KI), ACS grade.
d. Sodium hydroxide (NaOH), ACS grade.
e. Soluble starch.

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4.4.1.2 Solutions.

a. Sodium thiosulfate 1.00N.
b. Hydrochloric acid 1:1.
c. Alkaline potassium iodide (200 grams KI and 120 grams NaOH per liter).
d. Starch solution, 1 percent.

4.4.1.3 Procedure. Weigh 10.000 ± 0.001 g. of ground and mixed bleach sample into a dry 500-ml. Erlenmeyer flask containing a magnetic stirring bar. Add 200 ml. of distilled water and 50 ml. of alkaline potassium iodide solution. Agitate for at least 5 minutes to effect solution of the bleach, then slowly add 50 ml. of hydrochloric acid solution while continuing to agitate. (Rapid addition of the acid will cause too rapid release of the free iodine and presence of iodine vapors above the solution in the flask. Loss of iodine vapors to the atmosphere will give erroneous results.) Titrate immediately with thiosulfate solution to a straw color. Add 5 ml. of starch solution and continue titration slowly to the disappearance of blue color.

4.4.1.4 Calculation.

Percent available chlorine =

$$\frac{\text{ml. } Na_2S_2O_3 \times N \text{ of } Na_2S_2O_3 \times 0.03546 \times 100}{\text{Weight of sample (g)}}$$

4.4.2 Carbonates as CO_2. Carbonate content, expressed as CO_2, shall be determined in accordance with method 1401 of Fed. Test Method Std. No. 536 except that the fifth sentence under "Procedure" shall be modified to read as follows: "Weigh to the nearest 0.001 g. one hundred grams of the ground and mixed bleach sample, transfer the bleach to the Knorr flask, add 40 ml. of freshly boiled distilled water, add hydroxylamine sulfate (approximately equal in weight to the bleach sample - use flask large enough to accommodate sample and reagents and increase size of tube carrying acid into the sample flask as required), and close the apparatus with the entire train in place". Continue with the sixth sentence under "Procedure", except delete "20 milliliters" and substitute "40 milliliters" in sentence six.

4.4.3 Reaction with carbonate ion. Dissolve 2.00 ± 0.01 g. of ground and mixed bleach sample in 200 ml. of distilled water (at 25° ± 2°C.). Filter the solution to remove insoluble material. To 100 ml. of the filter solution add 2.00 ± 0.02 g. of anhydrous sodium carbonate (Na_2CO_3). Stir or shake the system for five minutes. Allow to stand for an additional five minutes. Compare this solution with the remaining filtered solution. Any visually observed

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evidence of a precipitate or a definite increase in turbidity shall constitute failure.

4.4.4 Determination of pH. Determine the pH at 25° ± 1°C. of a solution containing 200 parts per million available chlorine as prepared in 4.4.10.2 using a pH meter with an alkali-resistant glass electrode in accordance with ASTM E 70-68.

4.4.5 Determination of insoluble matter content. Weigh out 0.312 ± .001 g. of ground and mixed bleach sample. Place 500 ml. of distilled water in a 1000-ml. beaker and raise the temperature of the water to 65° ± 2°C. While stirring the water with a mechanical stirrer, add the weighed bleach into the beaker. Continue to stir slowly for 5 minutes. Filter the solution and precipitate, if any, into a tared Gooch or fritted glass crucible. Wash the insolubles with three 20-ml. portions of distilled water. Dry the insolubles in an oven at 105°C. to constant weight.

4.4.5.1 Calculation. Calculate percent weight of the insolubles as follows:

$$\text{Weight percent insolubles} = \frac{A - B}{C} \times 100$$

A = Weight of crucible plus insolubles.
B = Weight of crucible.
C = Weight of sample of laundry bleach.

4.4.6 Moisture content. The moisture content shall be determined in accordance with ASTM D 2022-64, sections 50 through 53. The sample designated in 4.3.4.2, ground and mixed, shall be used for the moisture determination.

4.4.7 Loss on ignition. Place one gram (1.000 ± .001) of ground and mixed bleach sample in a tared porcelain crucible. The crucible with its contents is set vertically in a triangle on a tripod and heated over a Meker-type burner for one hour. The crucible with its residual contents (after ignition) is cooled under desiccation and weighed.

4.4.7.1 Calculation.

$$\text{Percent loss on ignition} = \frac{(C + W - D)}{W} \times 100$$

C = Weight of crucible.
W = Weight of bleach sample
D = Weight of residue and crucible after ignition.

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4.4.8 Thermal decomposition point. A small Wood's metal bath shall be prepared with a 250-ml. metal beaker filled to approximately three-fourths capacity with molten Wood's metal. This bath shall be heated either over a Bunsen burner or in an electric beaker heating mantle with rheostat. A thermometer (0° to 360°C) shall be inserted in the Wood's metal one inch from the side of the beaker so the top of the mercury bulb is one inch below the surface. A safety shield shall then be placed in front of this assembly. The analyst should wear protective clothing and complete eye and face protection. Into a 1- by8-inch Pyrex test tube 5.0 grams of the unground product shall be placed. The bulb of a thermometer, 0° to 360°C., shall be immersed in this sample and the test tube shall be supported over the Wood's metal bath so that the top of the sample is just below the level of the Wood's metal. Heat shall be applied to the metal bath at such a rate that the temperature of the sample under test does not lag more than 10°C. below the temperature of the bath. The temperature of the bleach shall be watched constantly. Visual indications of rapid reaction such as mild explosion, fire or sudden rapid rise to a temperature greater than that of the bath shall be considered as evidence of thermal decomposition. The temperature taken as the point of decomposition is not the temperature reached after reaction has taken place but is rather the last observed temperature before the reaction.

4.4.9 Loss of available chlorine on heating. Place 5.0000 ± 0.0005 g. of unground bleach sample into each of two test tubes (150 mm. long and 25 mm. in diameter), and insert rubber stoppers carrying air condensers (glass tubes) 15 inches long and 1/4 inch in internal diameter. The glass tubes shall extend 1/2 inch beyond the stopper into the test tubes. Place the test tubes in a constant temperature bath at 80° ± 2°C. for 6 hours. At the end of the heating period, remove the tubes, substitute solid rubber stoppers for the air condenser stoppers, and allow to cool to room temperature. Determine the available chlorine by the method specified in 4.4.1 combining the two samples to perform this determination. In calculating the percent of available chlorine after heating, use the weight of the combined samples taken before heating.

4.4.9.1 Calculation.

Percent loss of available chlorine after heating =

$$\frac{(A - B) \times 100}{A}$$

Where: A = Percent available chlorine as received.
B = Percent available chlorine after heating.

4.4.10 Performance test.

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4.4.10.1 Preparation of stained swatches.

a. Cloth. The cloth shall be preshrunk, plain weave, permanent finish cotton and shall contain approximately 54 ends and 45 picks per square inch and weigh approximately 5.1 ounces per square yard. The initial reflectance of the cloth shall be determined using a Model 610 Photovolt or equivalent, and a green tri-stimulus filter. Thirty-two, 3.5- by 3.5-inch, swatches shall be prepared.

b. Staining. Bring one liter of tap water to a boil. Insert 5 tea bags (Orange Pekoe Cut Black) in the boiling water and boil for 5 minutes, then remove bags. Insert 32 cloth swatches (3.5 by 3.5 inches) and continue to boil for 5 minutes. Remove swatches, wring out and dry in a clothes dryer at 180° ± 10°F. for 30 minutes. Rinse in cold water and repeat drying procedure.

NOTE: Stained swatches older than 2 weeks shall be discarded and fresh swatches prepared. Record the reflectance of the stained swatches taking average readings of both sides. Use swatches stacked four deep for the readings in order to get a background of similar reflectance.

4.4.10.2 Stain removal.

a. Apparatus. A Launder-Ometer with pre-heating load table, or similar machine, shall be used, in which 1-pint glass jars are held with their bases toward a horizontal shaft and 2 inches from its center of rotation, the shaft having a speed of rotation of 40 to 45 r.p.m. The temperature of the bath shall be maintained at 49° ± 2°C.

b. Reagents.

Water. The hardness of the water shall not exceed 50 p.p.m. calculated as calcium carbonate ($CaCO_3$).

Detergent solution. A 0.4 percent solution of a synthetic detergent (bead form) conforming to type I of P-D-245 shall be used.

Bleach solution. A solution of laundry bleach shall be prepared in a 2-liter volumetric flask using sufficient ground and mixed dry bleach to give a concentration of 200 p.p.m. available chlorine (0.02 percent). Weigh bleach sample directly into the flask from a tared container of bleach and determine sample weight by difference. Prepare this solution immediately before use.

c. Procedure. Measure into each of five 1-pint glass jars, each containing ten 1/4-inch stainless steel balls, 150 ml. of detergent solution and 150 ml. of bleach solution both solutions at a temperature of 49°C. (120°F). Place two stained swatches in each jar. Cap the jars and place

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in washing machine. The machine is operated for 20 minutes at temperature of 49° + 2°C. After washing, the swatches are removed, rinsed thoroughly in water at a temperature of 45° - 50°C., squeezed by hand to remove excess water, smoothed flat, and air dried. Measure the reflectance of the bleach swatches, taking average readings on both sides. Use swatches stacked four deep for readings in order to get a background of similar reflectance.

4.4.10.3 Calculation. Calculate the total percent stain removed as follows:

$$\text{Total percent stain removed} = \frac{B - S}{O - S} \times 100$$

B = Average swatch reflectance after bleaching.
S = Average swatch reflectance of stained swatch.
O = Initial reflectance of original unstained swatch.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Two pound quantities. Two pounds of bleach shall be packaged in a folding paperboard box conforming to variety 2 style I, type A, class a or b of PPP-B-566. Each box shall be fitted with a 0.002-inch minimum thickness polyethylene bag conforming to type I or II, grade A, style 1 or 2 of PPP-B-26. The polyethylene bag may also be closed by double-folding the plastic material at the mouth and taping it to the bag body with 1-inch minimum width pressure-sensitive tape conforming to PPP-T-60 or PPP-T-76. Prior to or during the final closure operation excess air within the polyethylene bag shall be expelled. Each box shall be securely closed in accordance with the appendix of the box specification.

5.1.1.2 Twenty-five, 50 and 125-pound quantities. Twenty-five, 50 and 125-pound quantities of bleach shall not require packaging.

5.1.2 Level C. Bleach shall be packaged to afford adequate protection against physical damage and deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

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5.2.1.1 Two pound quantities. Twenty-four 2-pound packages of bleach, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of the container specification.

5.2.1.2 Twenty-five pound quantities. Twenty-five pounds of bleach shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, grade V2s of PPP-B-636. Each container shall be fitted with a 0.004-inch minimum thickness polyethylene bag conforming to type I or II, grade A, style 1 or 2 of PPP-B-26. The polyethylene bag may also be closed by double-folding the plastic material at the mouth and taping it to the bag body with 1-inch minimum width pressure-sensitive tape conforming to PPP-T-60 or PPP-T-76; or by tightly knotting the plastic material at the mouth; or tightly twisting the mouth opening, then folding over half of the twisted portion and applying a plastic coated wire tie around the center of the fold. Prior to or during the final closure operation, excess air within the polyethylene bag shall be expelled. Regardless of the closure method used, the polyethylene bag shall be of sufficient length to allow for a knotting closure. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of the container specification. When the bottom flaps are fastened with metal stitches or staples, the inside bottom of the container shall be provided with a fiberboard or chipboard pad. The length and width of the pad shall measure approximately 1/8-inch less than the corresponding length and width dimensions of the container.

5.2.1.3 Fifty and 125-pound quantities. Fifty and 125-pound quantities of bleach shall be packed in a snug-fitting fiber drum conforming to type III, grade A, class 1 of PPP-D-723. Each fiber drum shall be fitted with a 0.004-inch minimum thickness polyethylene bag conforming to type I or II, grade A, style 1 or 2 of PPP-B-26. The polyethylene bag shall be closed as specified in 5.2.1.2. Each fiber drum shall be closed in accordance with the applicable requirements of PPP-D-723.

5.2.2 Level B.

5.2.2.1 Two pound quantities. Twenty-four 2 pound packages of bleach packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF(variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.2 Twenty-five pound quantities. Twenty-five pounds of bleach shall be packed in a snug-fitting fiberboard shipping container conforming to style FOL, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each fiberboard container shall be fitted with a 0.004-inch minimum thickness

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polyethylene bag conforming to type I or II, grade A, style 1 or 2 of PPP-B-26. The polyethylene bag shall be closed as specified in 5.2.1.2. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification. When the bottom flaps or top flaps are fastened with metal stitches or staples, the inside bottom and top of the container (as applicable) shall be provided with a fiberboard or chipboard pad. The length and width of the pad shall measure approximately 1/8-inch less than the corresponding length and width dimensions of the container.

5.2.2.2.1 When specified (see 6.2), the fiberboard shipping container specified in 5.2.2.1 and 5.2.2.2 shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.2.3 Fifty and 125-pound quantities. Fifty and 125-pound quantities of bleach shall be packed in a snug-fitting fiber drum conforming to type I, grade A, class 1 of PPP-D-723. Each fiber drum shall be fitted with a 0.004-inch minimum thickness polyethylene bag conforming to type I or II, grade A, style 1 or 2 of PPP-B-26. The polyethylene bag shall be closed as specified in 5.2.1.2. Each fiber drum shall be closed in accordance with the applicable requirements of PPP-D-723. When specified (see 6.2) and alternatively, the interior of the drum shall be coated with a heavy coating of microcrystalline wax.

5.2.3 Level C. Bleach, packaged as specified in 5.1 (when applicable), shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Labeling. Unless otherwise specified in the contract or order, each interior package and shipping container shall be labeled in accordance with the Federal Hazardous Substances Act. The following legend shall also be included:

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DIRECTIONS FOR USE

Dry bleach may be used in white cotton wash formulas with either soap or sythetic detergents. During bleach operation add dry bleach to wash-wheel in the following quantities:

Average soil - 2 ounces per 100 lb. load.
Medium soil - 3 ounces per 100 lb. load.
Heavy soil - 4 ounces per 100 lb. load.
Keep container tightly closed.
Keep out of contact with combustible materials and laundry sours.
Store in cool, dry place.
Date of packaging.

5.3.3.1 Warning legend. Directly beneath "DIRECTIONS FOR USE" add, in red ink, the following legend:

WARNING

THIS PRODUCT IS NOT TO BE USED

FOR THE DISINFECTION OF WATER OR FRUITS OR VEGETABLES

6. NOTES

6.1 Intended use. The dry laundry bleach covered by this specification is intended for direct addition to the laundry wheel in laundering operations. This product is not intended to be used for the disinfection of water or fruits or vegetables.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Quantity put-up of bleach in interior packages and shipping containers (see section 5).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.2.1).
(e) When a wax coating is required for interior of the drum (see 5.2.2.3).

6.3 The laundry bleach must be stored in a cool, dry place. Stocks should be rotated and overstocking avoided. Not more than six month's supply should be purchased at one time. When not in use, the drums should be kept tightly closed.

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MILITARY CUSTODIANS:

Army - GL
Navy - SH,
Air Force - 68

Review activities:

Army - GL, MD, MU
Navy - SH, MS
Air Force - 68

User activity:

Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

AGR	JUS
COM-NBS	JUS-FPI
DC	VA
GSA	
HEW	

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stration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
O-B-420B BLEACH, LAUNDRY, ORGANIC CHLORINE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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